chhat छत

Solar, without the sales pitch.



The independent AI solar advisor for every Indian roof.

Team 404 not found HackIndia · AI-First Startup Hackathon 2026

Track: AI in any startup (climate) + FinTech

THE PROBLEM

India installs 16,000 rooftops a day. Nobody is on the homeowner's side.

51 lakh

PM Surya Ghar installations by Sep 2026; 1 crore homes targeted by Mar 2027

5%

of households willing to install, though the economics work (CEEW)

Jul 2026

MNRE acknowledges wrong DISCOM bills after rooftop installs

Every calculator a homeowner finds belongs to someone selling panels.

One quote. Three problems. ₹1.6 lakh at risk.

“3 kW” on-grid system	₹2,41,256
5 × 540 Wp, imported	2.70 kWp
GST @ 13.8%	₹29,256

- **₹57,710** above the fair price band
- **₹72,600** subsidy lost to non-DCR panels
- **0.3 kW** paid for and never installed
- Then after install: **₹2,714** overbilled in one cycle when the DISCOM ignores export

Four checks between a family and a bad deal.

Roof Report

Bill + 20 years of NASA sunlight → size, generation, subsidy, loan, 25-year payback.

Quote Check

14 checks: fair price, DCR and ALMM, capacity, inverter, warranties, GST, payment terms.

Bill Audit

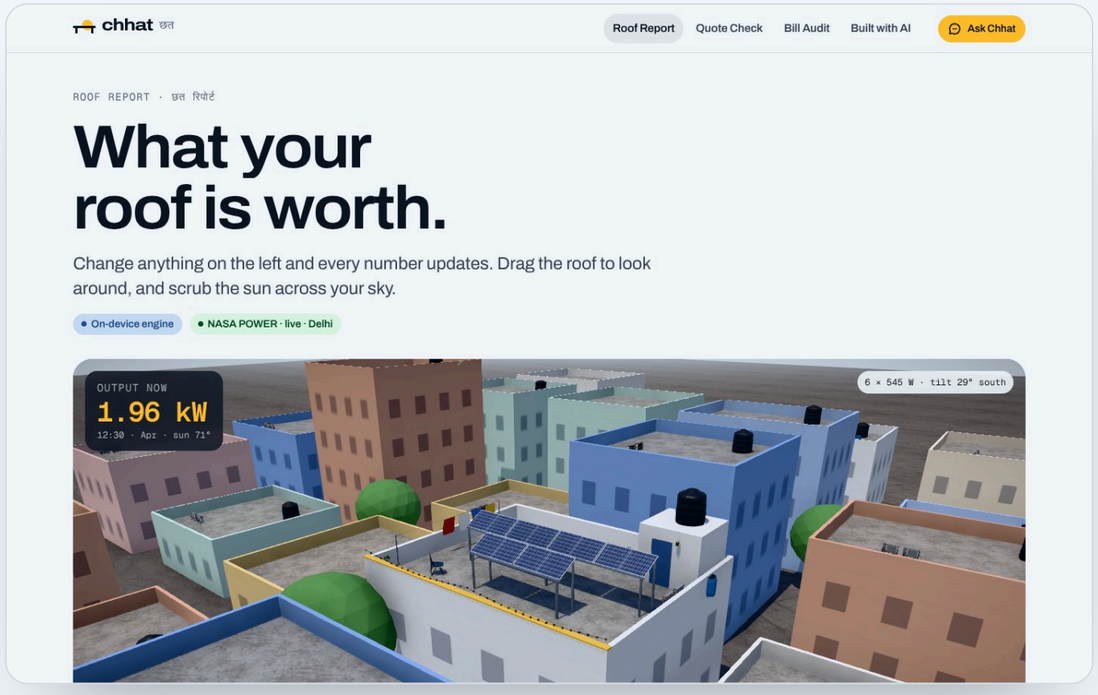
Import – export – credit vs. billed units. Drafts the DISCOM letter.

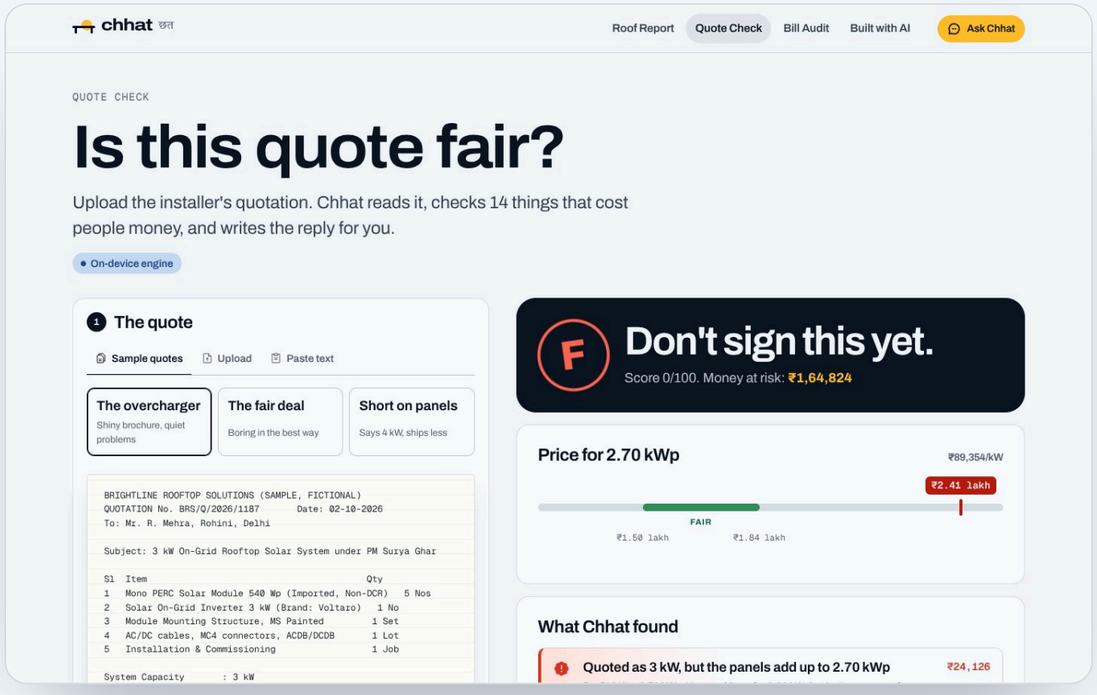
Ask Chhat

Agent in Hindi, Hinglish and English. Every number comes from a tool.

LIVE, NO LOGIN, NO KEY

Works for a stranger in ten seconds.





Claude decides. The engine computes. Nothing is invented.

- **7 typed tools**: irradiance, sizing, subsidy, finance, quote audit, bill audit, sourced knowledge base
- **Two agents, one toolset**: Claude Opus 5.5 on the server; an on-device router when there's no key or no network
- **Document reading**: Claude structured output, or on-device OCR and PDF text, always into an editable form
- **Real data**: NASA POWER (72 cities bundled, live for any lat/lon), MNRE rules, CEA emission factor

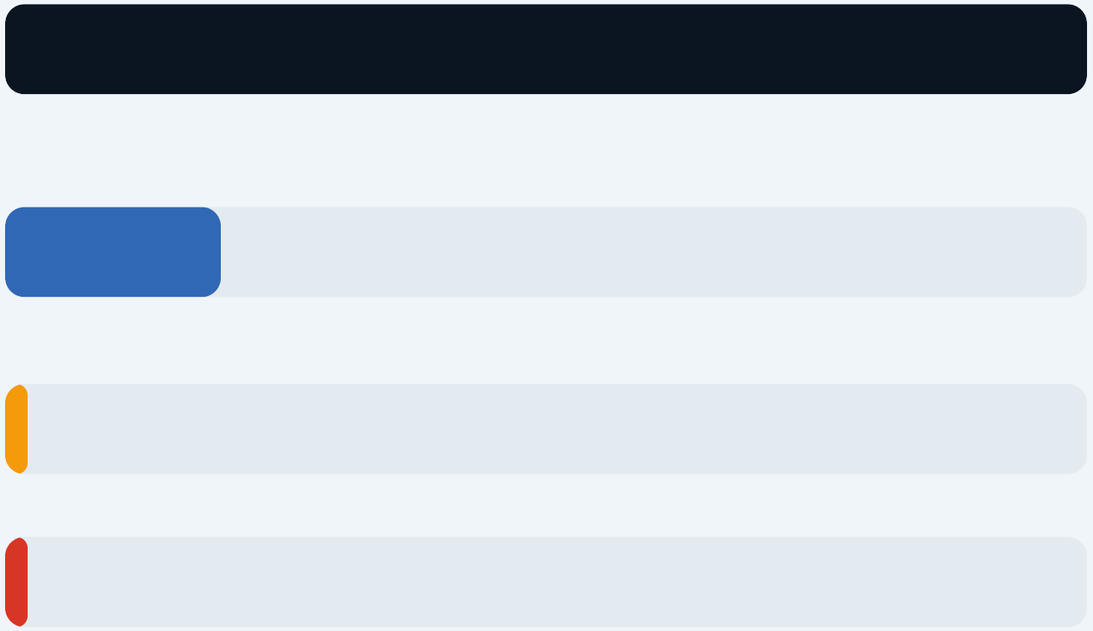
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"Vendor quoted 2.4 lakh for 3 kW in Lucknow, fair?"

→ audit_quote(system_kw: 3, total_price: 240000,
state_code: "UP")

← Grade C · ₹42,000 above fair range · DCR not
stated · no 5-yr warranty

"For 3 kW with DCR panels, a fair installed price is
₹1,62,000 to ₹1,98,000. This quote is about ₹42,000
above that."
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A ₹1.8 lakh crore installation market, refereed by no one.



₹1,82,400 cr

TAM · installation value of India's 32 GW economic residential rooftop potential (CEEW)

₹36,480 cr / yr

Planning base · 20 lakh homes a year after the scheme, a third of the 2026 peak

₹2,278 cr / yr

SAM · Chhat's fees on that flow, plus installer SaaS and state analytics

₹80 cr

SOM · FY31 revenue: 3.5% of SAM, 3% of national installs

BUSINESS MODEL

Free for homes. Paid by the people who want to sell to them.

STREAM	PAYER	PRICE	FY31 (₹ CR)
Verified leads	Installers	₹1,500 / lead	45.6
Success fee	Installers	2% of system value	17.8
Installer SaaS	Installers	₹2,999 / month	9.0
State / DISCOM analytics	Governments	₹40 L / state-year	4.0
Loan origination	Banks, NBFCs	1% of loan	3.4

UNIT ECONOMICS, FY28

₹4,803 contribution on every installed home.

₹196

revenue per report

₹96

variable cost per report (₹85 acquisition, ₹10 calls & payments, ₹2 AI)

2.0x

revenue to variable cost

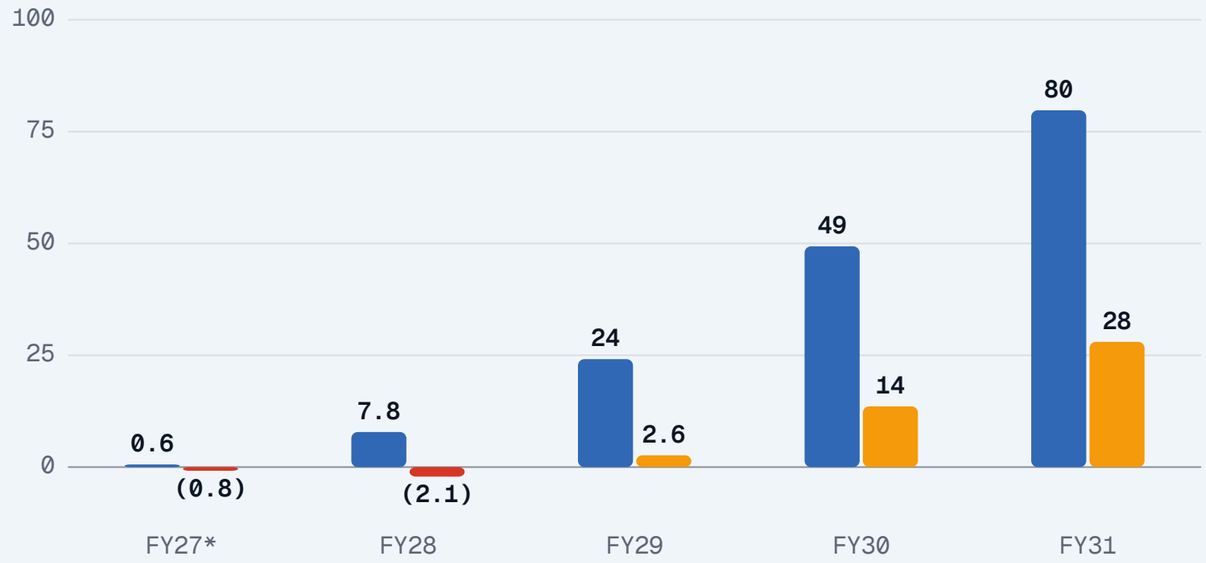
1.5%

of the home's 25-year savings is Chhat's take

Funnel: 8% of reports become verified leads; 20% of leads install. 62.5 reports per installed home. AI is under 1% of revenue.

FIVE-YEAR PLAN · ₹ CRORE

EBITDA positive in year three.



Revenue

EBITDA

FY31: ₹79.8 cr revenue, 35% EBITDA margin, ~124 people. Downside (30% less traffic): break-even Jan 2030, ~₹6 cr more needed. *FY27 is Nov to Mar.

A calculator is easy to copy. Our dataset isn't.

City price index

Every audited quote adds a dated price per kW, per city, per panel type.

DISCOM error map

Every audited bill shows which sub-divisions bill wrong. Regulators will pay for that.

Calibrated yield

Post-install exports tune the PV model city by city, beyond NASA averages.

COMPETITION

Installers sell panels. Chhat referees them.

	INDEPENDENT	AUDITS QUOTES	AUDITS BILLS	HINDI AGENT
SolarSquare, Freyr, Loom	✗	✗	✗	✗
PM Surya Ghar portal	✓	✗	✗	✗
Lead aggregators	✗	✗	✗	✗
Chhat	✓	✓	✓	✓

Start where the subsidy stacks highest.

Nov 2026	Live in Delhi NCR, UP, Maharashtra · first 20 installers
Mar 2027	40,000 reports · 3,200 verified leads · first lender
Sep 2027	10 states · “Chhat Verified” badge · first DISCOM pilot
FY28	₹7.8 crore revenue · 4,900 installs
FY29	EBITDA positive from Apr 2028 · optional Series A for C&I and batteries

Growth loop: every report makes a shareable roof card for WhatsApp.

AI-FIRST EXECUTION

Two briefs in. A startup out.

16,008

lines of source, all written by Claude Code

124

automated tests: unit, plus end-to-end in a real browser

25

ideas generated, debated and scored before writing code

0

lines of code written by a human

Research, idea tournament, plan, PV model, 3D world, copy, financial model, this deck: AI. The two briefs and the review: human. Receipts at /built-with-ai.

THE ASK

₹6 crore seed for 15%.

- **38%** growth
- **34%** product & engineering
- **18%** G&A, compliance & state sales
- **10%** installer network & lead verification

Gets us to EBITDA break-even in April 2028 with ₹1.9 crore of cash at the low point.

Try it: read your roof at the live link. Every number cites its source.